

3.2 Hydrocarbons

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.2 Hydrocarbons
Difficulty	Hard

Time allowed: 50

Score: /35

Percentage: /100

Question 1a

This question is about free radical substitution.

1,2-dibromoethane reacts with bromine in UV light to produce a mixture of further substituted haloalkanes.

i)

Write an equation for the initiation step.

[1]

ii)

Explain why this is an example of homolytic fission.

[1]

[2 marks]

Question 1b

Write **two** equations showing the propagation of this chain reaction to produce 1,1,2-tribromoethane.

[2 marks]

Question 1c

Traces of 1,2,3,4-tetrabromobutane are found in the reaction mixture.

i)

Write an equation to show how this product is formed.

[1]

ii)

Write a balanced symbol equation to show the overall reaction between 1,1,2-tribromoethane with bromine in UV light to form hexabromoethane.

[1]

[2 marks]

Question 1d

Using the information from this question and your own knowledge, suggest the limitations of free radical substitution.

[2 marks]

Question 2a

This question is about hydrocarbons.

10 cm^3 of a gaseous hydrocarbon, C_xH_y , was reacted with 100 cm^3 of oxygen gas, an excess. The final volume of the gaseous mixture was 95 cm^3 .

This gaseous mixture was treated with concentrated, aqueous sodium hydroxide to absorb the carbon dioxide present. This reduced the gas volume to 75 cm^3 .

All gas volumes were measured at 298 K and 100 kPa .

i)

Write an equation for the reaction between sodium hydroxide and carbon dioxide.

[1]

ii)

Calculate the volume of carbon dioxide produced by the combustion of the hydrocarbon.

volume of CO_2 produced = cm^3
[1]

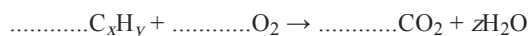
iii)

Calculate the volume of oxygen used up in the reaction with the hydrocarbon.

volume of O_2 used = cm^3
[1]

iv)

Use your answers to **(b)(ii)** and **(b)(iii)**, together with the initial volume of hydrocarbon, to balance the equation below.



[2]

v)

Deduce the values of x , y and z in the equation in **(iv)**.

$x = \text{.....}$

$y = \text{.....}$

$z = \text{.....}$

[3]

[8 marks]

Question 2b

A branched-chain hydrocarbon, **W**, with the formula C_4H_8 , reacts with hydrogen bromide, HBr , to give two products **X** and **Y**.

X and **Y** are structural isomers of molecular formula C_4H_9Br . More **X** is produced than **Y**.

i)

Give the structural formulae **and** names of the compounds **W**, **X**, and **Y**.

Name of **W**

Structural formula of **W**

Name of **X**

Structural formula of **X**

Name of **Y**

Structural formula of **Y**

[3]

ii)

When **W** reacts with hydrogen bromide, more **X** than **Y** is produced. Explain why.

[2]

[5 marks]

Question 2c

Pent-2-ene will react with HBr.

Draw the mechanism for the reaction of pent-2-ene with hydrogen bromide. You do not need to include any partial charges.

[2 marks]

Question 3a

A student completed an investigation into the amount of C=C bonds in a molecule of cooking oil, with a density of 0.526 g cm^{-3} , by titrating samples with 0.02 mol dm^{-3} bromine water.

The student plans to use the following procedure for their investigation.

1. Measure 5.0 cm^3 of solvent into a conical flask.
2. Add five drops of cooking oil to the solvent, using a dropping pipette.
3. Swirl to ensure mixing.
4. Note the initial reading of the bromine water in the burette.
5. Add the $2.0 \times 10^{-2} \text{ mol dm}^{-3}$ bromine water from the burette to the solution in the conical flask slowly.
 - Shake vigorously after each addition until the bromine colour disappears.
6. As the bromine colour takes longer to fade with each addition, add less bromine water each time until there is just an excess of bromine in the flask.
 - This will be shown by a permanent yellow tint.
7. Note the final reading of the burette.

Suggest **two** improvements that the student could make to this procedure to obtain more consistent results.

[2 marks]

Question 3b

The student's titration results are shown in Table 3.1.

Table 3.1

Experiment	Initial volume / cm ³	Final volume / cm ³
Rough	0.00	26.35
1	0.00	26.25
2	0.10	26.45
3	0.00	26.10

The approximate M_r of the cooking oil is 1002 and its density is 0.526 g cm^{-3} .

Assume each drop of cooking oil has a volume of $5.0 \times 10^{-2} \text{ cm}^3$.

The concentration of bromine water used was $2.0 \times 10^{-2} \text{ mol dm}^{-3}$.

Use this data and the titration results from Table 3.1 to deduce the number of C=C double bonds in a molecule of the oil. Show your working.

[6 marks]

Question 3c

Fig. 3.1 shows the structure of a mono-unsaturated oil and a polyunsaturated oil.

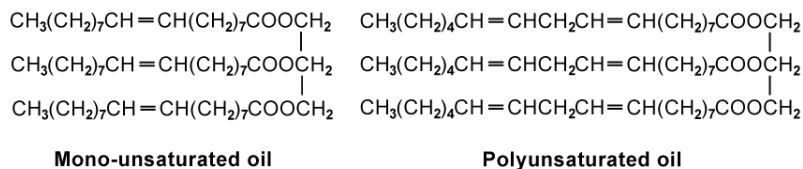


Fig. 3.1

- i)
Suggest what is meant by the terms mono-unsaturated oil and polyunsaturated.
- ii)
Suggest the reagent and conditions necessary to produce the mono-unsaturated oil from the polyunsaturated oil.

[2]

[2]

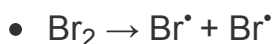
[4 marks]

Model Answers: Hard

1a

a)

i) The initiation step for this is:



OR



ii) This is an example of homolytic fission because:

- Each bromine atom receives one electron from the covalent bond / shared pair of electrons; [1 mark]

[Total: 2 marks]

- The initiation step in free radical substitution is always the same, the only thing that changes is the halogen that might be present
- There are two types of fission:
 - Homolytic whereby each atom takes one electron from the covalent bond
 - Heterolytic whereby one atom takes both electrons from the covalent bond
- The type of fission which occurs depends on the difference in electronegativity of the atoms
- Br_2 consists of two bromine atoms which obviously have the same electronegativity and so the electrons are split equally
- HBr would undergo heterolytic fission due to bromine being more electronegative than hydrogen

1b

b) The propagation steps are:





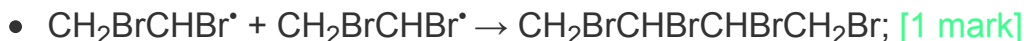
[Total: 2 marks]

- In the first step, the bromine free radical reacts with the organic molecule and extracts a hydrogen atom which forms HBr
 - The organic molecule is left with an unpaired electron from where the hydrogen atom was removed
- In the second step, the organic free radical reacts with another molecule of bromine to replace the lost hydrogen
 - As a result, another bromine free radical is formed
- **Remember:** Your equations for propagation steps should always have a molecule and a free radical as reactants and a different molecule and free radical as products
- It is not propagation if there are two molecules on one side of the equation

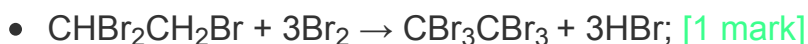
1c

c)

i) The equation to form 1,2,3,4-tetrabromobutane is:



ii) The balanced symbol equation for the overall reaction between 1,1,2-tribromoethane with bromine to form hexabromoethane is:



[Total: 2 marks]

- The step to form 1,2,3,4-tetrabromobutane is a termination step in the mechanism
- During termination two free radicals join together
- The two free radicals that form $\text{CH}_2\text{BrCHBrCHBrCH}_2\text{Br}$ are two lots of $\text{CH}_2\text{BrCHBr}^\bullet$
- Some students find it helpful to draw the displayed formula of the radicals and the final product
- All reactant and product names are given to you should it should be pretty straightforward to write the equation for these

1d

d) The limitations of free radical substitution are:

Any **two** of the following:

- Further substitution can occur; [1 mark]
- The mixture can contain impurities; [1 mark]
- The halogen can be substituted at different positions in the carbon chain; [1 mark]
- The reaction is difficult to control [1 mark]

[Total: 2 marks]

- Two examples of limitations are presented to you throughout other parts of the question:
 - Part (c) tells you that traces of 1,2,3,4-tetrabromobutane are found in the mixture which would be classed as an impurity formed from the organic free radicals
 - Part (c) also asks for an overall reaction to form hexabromoethane
 - Hexabromoethane would be formed via further substitution
- Depending on the chain length, a mixture of different monosubstituted isomers can be formed
 - For example, propane could form 1-bromopropane and 2-bromopropane

2a

a)

i) An equation for the reaction between sodium hydroxide and carbon dioxide is:

- $2\text{NaOH} + \text{CO}_2 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$; [1 mark]

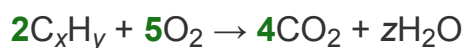
ii) The volume of carbon dioxide produced by the combustion of the hydrocarbon is:

- $95 - 75 = 20 \text{ (cm}^3\text{)}$; [1 mark]

iii) The volume of oxygen used up in the reaction with the hydrocarbon is:

- Excess oxygen - 75 cm^3 so used = $25 \text{ (cm}^3\text{)}$; [1 mark]

iv) The balanced equation is:



- All balancing numbers correct; [2 marks]
- 2 balancing numbers correct; [1 mark]

v) The values of x , y and z in the equation in (iv) are:

- $x = 2$; [1 mark]
- $y = 2$; [1 mark]
- $z = 2$; [1 mark]

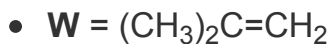
[Total: 8 marks]

- The final gaseous mixture (volume 95 cm^3) contains oxygen (as it was in excess) and carbon dioxide
- Once the carbon dioxide had been removed, this volume reduced to 75 cm^3 , so the volume of carbon dioxide produced can be calculated by $95 - 75 = 20 \text{ cm}^3$
- This means that 75 cm^3 of oxygen was left and initially, there was 100 cm^3 of oxygen, therefore 25 cm^3 must have been used
- The question also tells you that 10 cm^3 of hydrocarbon was used
- The ratio of the volume of hydrocarbon : volume of oxygen : volume of carbon dioxide is $10 : 25 : 20$, simplified this gives $2 : 5 : 4$
 - This gives the molar ratio of hydrocarbon : oxygen : carbon dioxide as $2 : 5 : 4$
 - So the equation is $2\text{C}_x\text{H}_y + 5\text{O}_2 \rightarrow 4\text{CO}_2 + z\text{H}_2\text{O}$
- There are 4 carbons on the right, so in order for there to be 4 carbons on the left, x must be 2
 - $2\text{C}_2\text{H}_y + 5\text{O}_2 \rightarrow 4\text{CO}_2 + z\text{H}_2\text{O}$
- There are 10 oxygen atoms on the left, so in order for there to be 10 oxygen on the left, z must be 2 (there are already 8 oxygen atoms from CO_2 , so the remaining 2 must come from H_2O)
 - $2\text{C}_2\text{H}_y + 5\text{O}_2 \rightarrow 4\text{CO}_2 + 2\text{H}_2\text{O}$
- There are 4 hydrogen atoms on the right, so in order for there to be 4 hydrogens on the left, y must be 2
 - $2\text{C}_2\text{H}_2 + 5\text{O}_2 \rightarrow 4\text{CO}_2 + 2\text{H}_2\text{O}$

2b

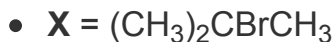
b)

i) The structures **and** names of the compounds **W**, **X**, and **Y** are:



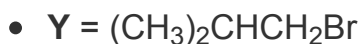
AND

2-methylpropene; [1 mark]



AND

2-bromo-2-methylpropane; [1 mark]



AND

1-bromo-2-methylpropane; [1 mark]

ii) When **W** reacts with hydrogen bromide, more **X** than **Y** is produced because:

Any **two** of the following:

- Halide bonds to the carbon with the highest number of alkyl groups / most substituted carbon; [1 mark]
- (Because) a tertiary carbocation is more stable than a primary cation; [1 mark]
- Due to the inductive effect of three alkyl groups; [1 mark]

[Total: 5 marks]

- 2-methylpropene is the only possibility of a branched chain isomer with the formula

C_4H_8

- Once the carbon dioxide had been removed, this volume reduced to 75 cm^3 , so the volume of carbon dioxide produced can be calculated by $95 - 75 = 20 \text{ cm}^3$
- This means that 75 cm^3 of oxygen was left and initially, there was 100 cm^3 of oxygen, therefore 25 cm^3 must have been used
- The question also tells you that 10 cm^3 of hydrocarbon was used
- The ratio of the volume of hydrocarbon : volume of oxygen : volume of carbon dioxide is $10 : 25 : 20$, simplified this gives $2 : 5 : 4$
 - This gives the molar ratio of hydrocarbon : oxygen : carbon dioxide as $2 : 5 : 4$
 - So the equation is $2\text{C}_x\text{H}_y + 5\text{O}_2 \rightarrow 4\text{CO}_2 + z\text{H}_2\text{O}$
- There are 4 carbons on the right, so in order for there to be 4 carbons on the left, x must be 2
 - $2\text{C}_2\text{H}_y + 5\text{O}_2 \rightarrow 4\text{CO}_2 + z\text{H}_2\text{O}$
- There are 10 oxygen atoms on the left, so in order for there to be 10 oxygen on the left, z must be 2 (there are already 8 oxygen atoms from CO_2 , so the remaining 2 must come from H_2O)
 - $2\text{C}_2\text{H}_y + 5\text{O}_2 \rightarrow 4\text{CO}_2 + 2\text{H}_2\text{O}$
- There are 4 hydrogen atoms on the right, so in order for there to be 4 hydrogens on the left, y must be 2
 - $2\text{C}_2\text{H}_2 + 5\text{O}_2 \rightarrow 4\text{CO}_2 + 2\text{H}_2\text{O}$

2b

b)

i) The structures **and** names of the compounds **W**, **X**, and **Y** are:

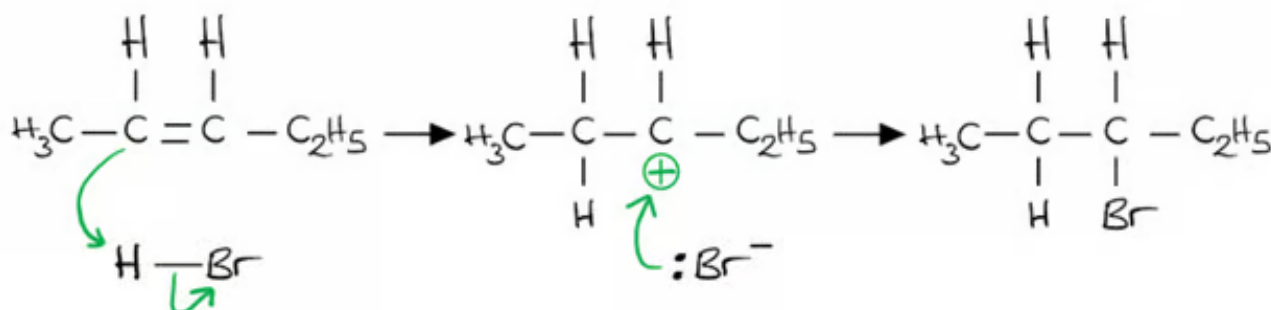
- **W** = $(\text{CH}_3)_2\text{C}=\text{CH}_2$
AND
2-methylpropene; [1 mark]
- **X** = $(\text{CH}_3)_2\text{CBrCH}_3$
AND
2-bromo-2-methylpropane; [1 mark]
- **Y** = $(\text{CH}_3)_2\text{CHCH}_2\text{Br}$
AND
1-bromo-2-methylpropane; [1 mark]

ii) When **W** reacts with hydrogen bromide, more **X** than **Y** is produced because:

- Halide bonds to the carbon with the highest number of alkyl groups / most substituted carbon; [1 mark]
- (Because) a tertiary carbocation is more stable than a primary cation; [1 mark]
- Due to the inductive effect of three alkyl groups; [1 mark]

- 2-methylpropene is the only possibility of a branched chain isomer with the formula C_4H_8
- When it reacts with HBr, the δ^+ H of H-Br acts as an electrophile and accepts a pair of electrons from the C=C bond
- This forms a carbocation intermediate with the positive charge either on the carbon in position 2 or position 1
 - The carbon in position 2 is attached to 3 alkyl groups so it is a tertiary carbocation
 - The carbon in position 1 is attached to 1 alkyl group so it is a primary carbocation
- The carbocation with the positive charge on the carbon in position 2 is more stable as the alkyl groups push electron density towards the carbocation
- The bromide ion reacts with the carbocation
- As the tertiary carbocation is more stable, the major product will have bromine in position 2

c) The mechanism for the reaction of pent-2-ene with hydrogen bromide is:



- Curly arrow from the C=C bond to the H atom (in the H-Br molecule); [1 mark]
- Curly arrow from the H-Br bond to the Br atom; [1 mark]

- Correct carbocation intermediate drawn; [1 mark]
- Curly arrow from the lone pair on the :Br⁻ to the C⁺ of the carbocation; [1 mark]

[Total: 4 marks]

- The electrophile (in this case the slightly positively charged H atom in H-X) is attracted to the high electron density of the C=C bond
 - The electrophile then accepts a pair of electrons from the double bond
 - This results in the formation of a carbocation (positively charged carbon)
- The Br⁻ ion forms a bond with the positively charged carbon of the carbocation intermediate
- This question says that you do not need to include any partial charges
 - But, unless this is stated you should include the δ^+ on the H and the δ^- on the X of the H-X molecule

3a

a) Two improvements that the student could make to this procedure to obtain more consistent results are:

Any **two** of the following:

- Weigh out the oil (as its density is known); [1 mark]
- Use a measured / known volume of cooking oil; [1 mark]
- Make a standard solution of the cooking oil and solvent; [1 mark]
- Titrate the bromine water against 25.0 cm³ samples (of the cooking oil / solvent mixture); [1 mark]

[Total: 2 marks]

- You should recognise that scientific methods, such as titrations, work with known volumes and concentrations to identify one unknown quantity
- You have the concentration of the bromine water ($2.0 \times 10^{-2} \text{ mol dm}^{-3}$) and the volume from the titres
- You will want to work out moles or concentration of the cooking oil so you need to know its volume but 5 drops is not precise
- This means that you need to ensure that the “amount” of cooking oil used is consistent as a small volume, e.g. 5 drops, will have a high percentage error

- You can achieve this by either:
 - Measuring out a specific volume using a graduated / volumetric pipette
 - Weighing out the mass of cooking oil used as you know the density of the cooking oil

3b

b) To deduce the number of C=C bonds in the cooking oil:

- Mass of oil = density x vol of 5 drops = $0.526 \times (5 \times 10^{-2} \times 5) = 0.1315 \text{ (g)}$; [1 mark]
- Moles of oil = $\frac{0.1315}{1002} = 1.312 \times 10^{-4} \text{ moles}$; [1 mark]
- Average titre = $\frac{26.25 + 26.35}{2} = 26.30 \text{ cm}^3$; [1 mark]
- Moles of $\text{Br}_2 = 2.0 \times 10^{-2} \times \frac{26.30}{1000} = 5.26 \times 10^{-4} \text{ moles}$; [1 mark]
- Ratio of cooking oil : bromine = $1.312 \times 10^{-4} : 5.26 \times 10^{-4} = 1 : 4$; [1 mark]
- Therefore, there are 4 C=C bonds in the cooking oil; [1 mark]

[Total: 6 marks]

- You have all of the information that you need to calculate the moles of cooking oil and the moles of bromine
 - For the cooking oil you have the mass and density
 - For the bromine, you have the concentration and the average titre
 - **Remember:** Rough titres are **not** used in titration calculations, even if they are concordant
 - You would **not** be awarded the mark if you used the rough titre value in the calculation to find the average titre, i.e. your calculation for the average titre was $\frac{26.35 + 26.25 + 26.35}{3} = 26.32 \text{ cm}^3$
- The real question here is what you do once you have the moles of cooking oil and bromine
 - Theoretically, one mole of C=C in the cooking oil will react with one mole of bromine
 - The molar ratio of cooking oil to bromine tells you that 4 times the amount of bromine reacts with cooking oil
 - This implies that there are 4 C=C in the cooking oil

3c

c)

i) The terms mean:

- Mono-unsaturated: each carbon chain contains one C=C bond / carbon-carbon double bond; [1 mark]
- Polyunsaturated: each carbon chain contains two or more C=C bonds / carbon-carbon double bonds

OR

Each carbon chain contains many C=C bonds / carbon-carbon double bonds; [1 mark]

ii) The reagent and conditions necessary to produce the mono-unsaturated oil from the polyunsaturated oil are:

- Reagent: H₂ / hydrogen; [1 mark]
- Conditions: Pt / Ni catalyst

AND

Heat; [1 mark]

[Total: 4 marks]

